Motor Sizing

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Motor Sizing

- **Motor:** Converts electrical energy into mechanical energy. In this course we will be considering motors which rotate (spin)

- **Mechanical Power:** The mechanical output of the motor (P) measured in Watts.

- Older motors and USAmotors measure power in Horse Power(Hp).

- Conversion Factor: 1 Hp = 746 W

Motor Sizing

- Speed: The number of revolutions per minute of the motor. Usually denoted “N” and measures in rpm (revs per minute). In some formulae speed is measured in revs per second and denoted “n”. In academic works a different units called radians per second is often used and denoted “ω.

- Conversion Factors:

$$n=N/60$$

$$\omega=2\pi N/60$$

- Torque: The force with which a motor turns is called Torque measured in Newton meters (Nm). Amotor with high Torque accelerates quickly and can drive a heavy load.

Motor Sizing

- Relationship between Power, Speed and Torque

$$P = \frac{2 \times \pi \times T \times N}{60}$$

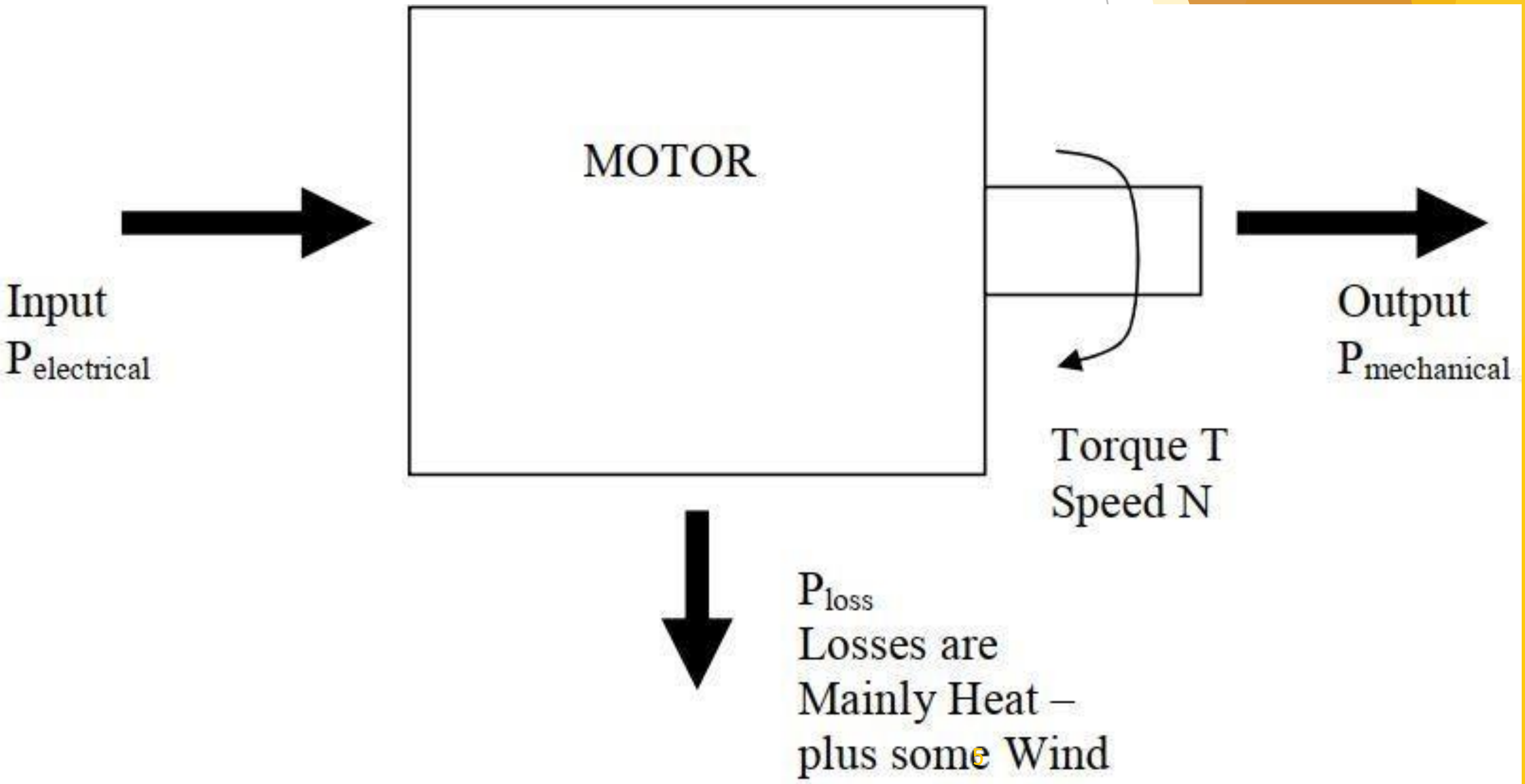
$$RMS\,power = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + \ldots\ldots + P_n^2 t_n}{t_1 + t_2 + t_3 + \ldots\ldots + t_n}}$$

Motor Sizing

- Losses and Efficiency

$$Input = Output + Losses$$

$$P_{electrical} = P_{mechanical} + P_{loss}$$



$$Efficiency = \eta = \frac{Output}{Input} \times 100\%$$

$$\eta = \frac{P_{mechanical}}{P_{electrical}} \times 100\%$$

Motor Sizing

Motor sizing refers to the process of picking the correct motor for a given load. It is important to size a motor correctly because:

- If a motor is too small for an application it may not have sufficient torque to start the load and run it up to the correct speed. Even if it does get the load up to speed the motor will overheat and burnout if it is too small for the application.

- If a motor is too large for an application then money has been wasted in purchasing such a large motor. Also motors typically operate inefficiently when they are run well below rated load. So money is also wasted in running costs.

Choosing The Right Motor

- For the load torque is constant then choosing a motor whose rated torque is slightly above the torque required by the load. The load torque should be between 75%-100% of the rated motor toque with 95% being an ideal choice.

- For constant speed applications where the motor will run continuously at rated speed you do not need to work out torque. You can simply look at load power and motor power and use the same 75%-100% rule. This is the case for standard induction motors without a variable speed drive.

Can The Motor Start The Load?

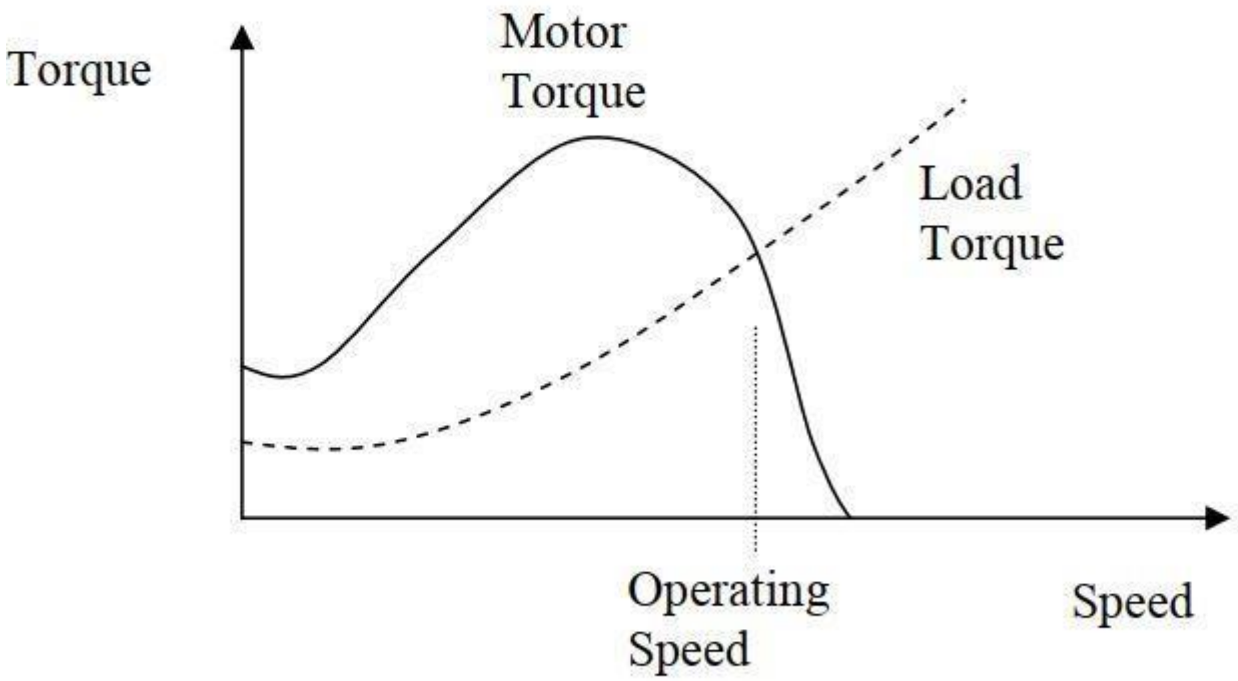
- The torque produced by a motor varies with speed and the torque produced by a load also varies with speed.

- If the motor torque is greater than the load torque then the load will accelerate.

- If the load torque is greater than the motor torque then the load will decelerate.

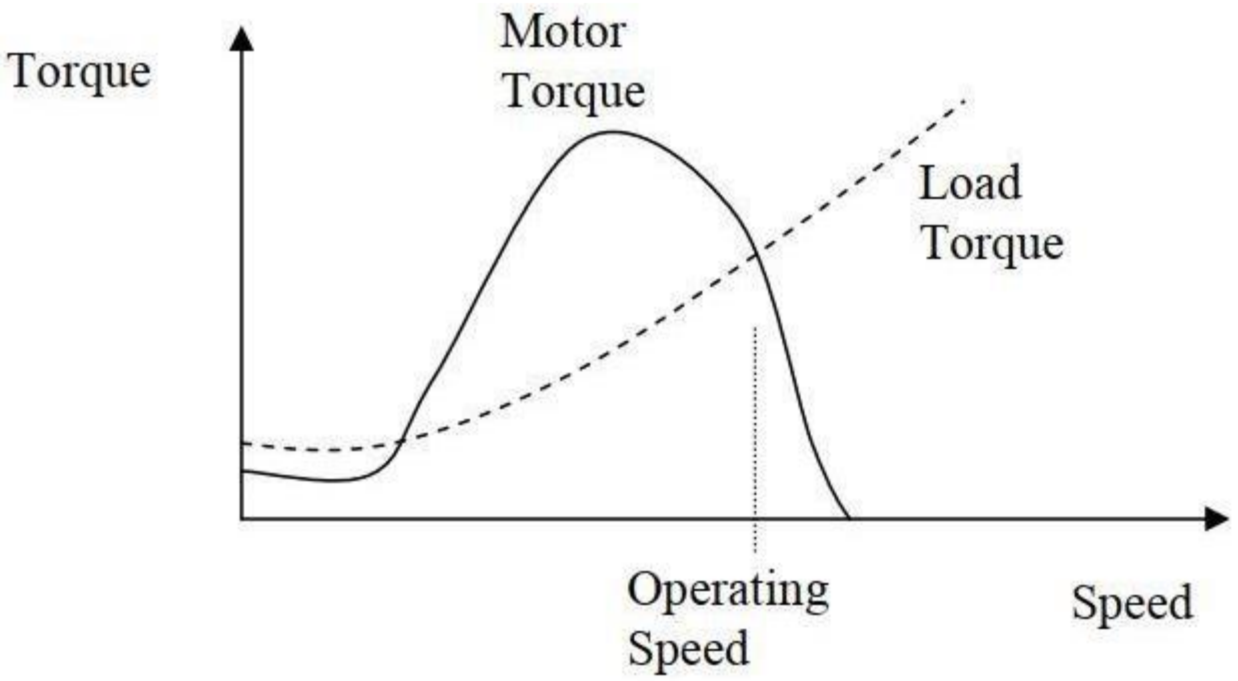
$$\text{Accelerating Torque} = T_{motor} - T_{load}$$

Can the Motor Start with the Load?



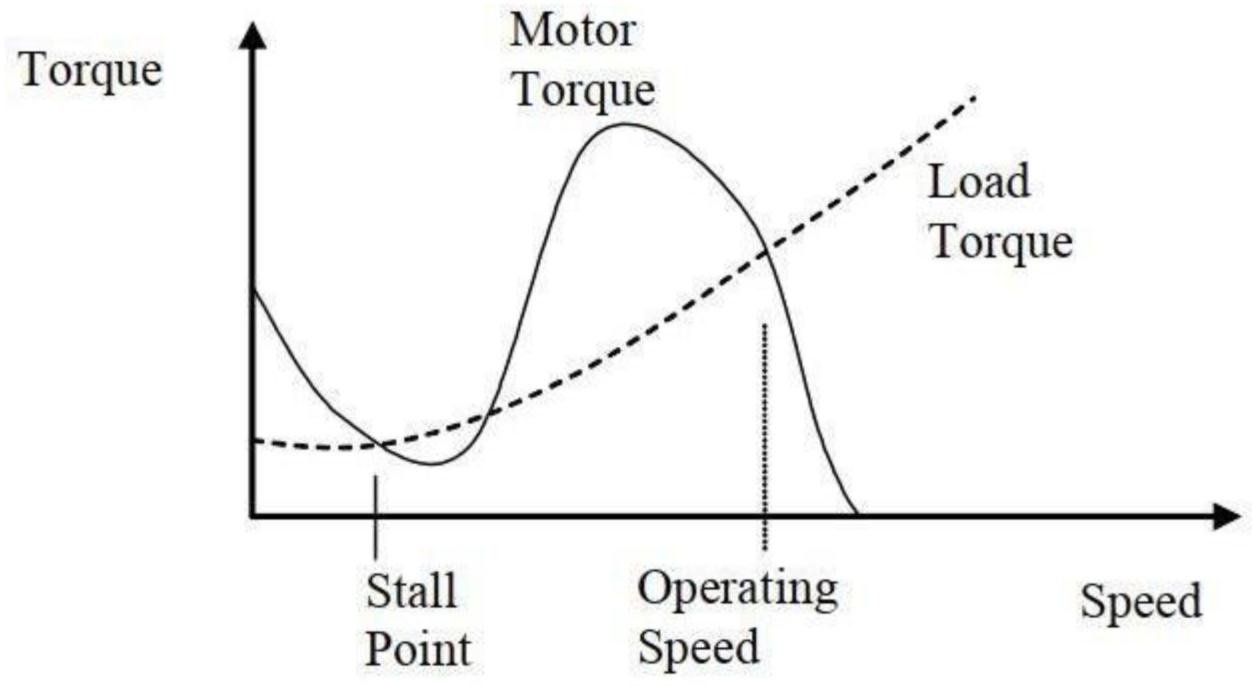
This motor will start the load and get it up to speed correctly.

Can the Motor Start with the Load?



This motor will never even start

Can the Motor Start with the Load?



This motor will start the load but it will get stuck at the stall point.

What If The Load Varies?

Some loads do not present a constant torque even after they have got up to full speed. This presents a variable power to the motor and complicates the sizing problem.

In general we should ensure that

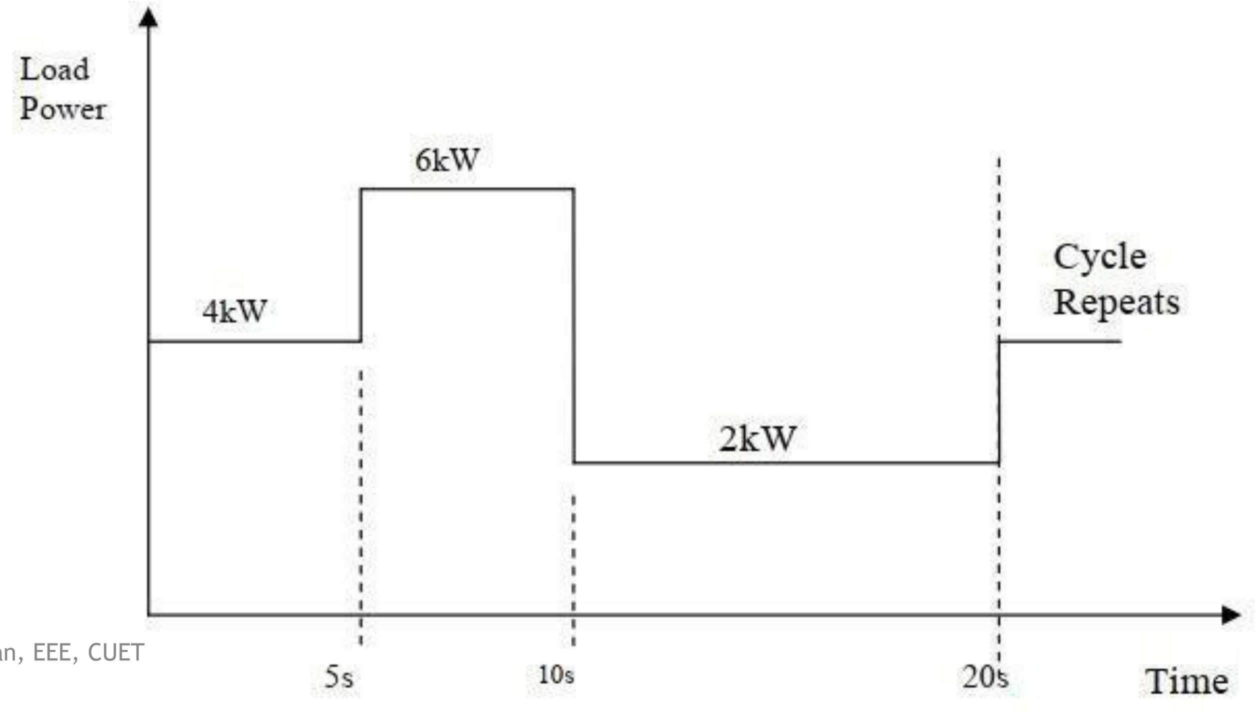
1. Peak load torque (or power) < Rated Motor torque (or power) x (1+Service factor/100%)

2. The root mean square load torque (power) requirement must be less than 100% of the rated motor torque (power) and ideally greater than 75% of rated motor torque (power).

3. Check that the Motor can start the load and get it up to speed.

Example

The following load torque profile was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 20%. You do not need to consider starting performance.



Solution

Sizing the motor:

- Peak Load Power requirement = 6kW.

Peak load power must be less than Rated Motor Power x Service Factor

$$=> 6kW < P_{motor} \times 1.2$$

$$=> P_{motor} > 6/1.2 = 5kW$$

- The RMS load power should ideally be between 75% and 100% of Rated Motor Power.

$$RMS Load power = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + \ldots + P_n^2 t_n}{t_1 + t_2 + t_3 + \ldots + t_n}} = \sqrt{\frac{4^2 \times 5 + 6^2 \times 5 + 2^2 \times 10}{5 + 5 + 10}} = 4.16kW$$

Solution

Sizing the motor:

- If we choose P_{motor}= 5kW then RMS load power = 4.16/5*100% = 83% of P_{motor} which is acceptable.

NB: Important If RMS load power turned out to be less than 75% of 5kW we sould still need to choose a 5kW motor in order to be able to supply the peak power. The RMS load power should ideally be between 75% and 100% of Rated Motor Power requirement. However if RMS load power was great than 100% of rated motor power we would need to choose a bigger motor to meet this requirement.

- Starting Performance – Not a requirement in this case:

Answer: Choose a 5kW motor.